

MAT111 Take-Home Mid

Navid Tawsif, 1000058969

November 30, 2025

Problem 1

3600

Problem 2

145

Problem 3

$$\begin{aligned}\Gamma(1) &= \int_0^\infty t^0 e^{-t} dt = [-e^{-t}]_0^\infty = 1 = (1-1)! \\ \Gamma(2) &= \int_0^\infty t e^{-t} dt = [-te^{-t}]_0^\infty + \Gamma(1) = 1 = (2-1)! \\ \Gamma(3) &= \int_0^\infty t^2 e^{-t} dt = [-t^2 e^{-t}]_0^\infty + 2\Gamma(2) = 2 = (3-1)! \\ \Gamma(4) &= \int_0^\infty t^3 e^{-t} dt = [-t^3 e^{-t}]_0^\infty + 3\Gamma(3) = 6 = (4-1)!\end{aligned}$$

Problem 4

Construct the great circle with any plane containing 2 of the given 5 points and the center of the sphere. Hence, both hemispheres contain these 2 points. 3 points remain, but there are only 2 hemispheres. By pigeonhole principle, one of the hemispheres contains at least $2 + \lceil \frac{3}{2} \rceil = 4$ of the points.

Problem 5

Assuming an “Either __ or __” sentence uses an exclusive or, i.e. exactly 1 of the statements to the left or right of ‘or’ is true.

- (a) The crime took place in the billiard room $\rightarrow$ The weapon was not the wrench $\rightarrow$ Professor Plum is innocent $\rightarrow$ Colonel Mustard is guilty.
- (b) The crime did not take place in the billiard room $\rightarrow$ The weapon was the wrench $\rightarrow$ Professor Plum is guilty.
- (c) The crime was committed with a knife $\rightarrow$ The weapon was not the wrench $\rightarrow$ Professor Plum is innocent $\rightarrow$ Colonel Mustard is guilty.
- (d) The crime took place at midnight $\rightarrow$ Colonel Mustard is guilty.

Problem 6

234

Problem 7

30

Problem 8

455

Problem 9

$16!/2^8$

Problem 10

415

Problem 11

$1/99$

Problem 12

Let $S = \{1, 2, 3, \dots, 2n\}$. Let $R = \{(x, y) \in S \times S : x + y = 2n + 1, x \leq y\}$ be the set of pairs from S whose sum equals $2n + 1$. By inspection,

$$\begin{aligned}1 + 2n &= 2n + 1 \\2 + 2n - 1 &= 2n + 1 \\&\vdots \\n + n + 1 &= 2n + 1\end{aligned}$$

we find that $|R| = n$. But we have to select $n + 1$ numbers from S , we must select both numbers of at least one pair.

Problem 13

Let $S = \{2m + 1 \in [1, 3n]\}$. Let $R = \{(x, y) \in S \times S : x/y = 3^z, z \in \mathbb{Z}\}$ be an equivalence relation on S . Definitely, selecting two numbers from the same equivalence class will result in the smaller dividing the larger.

To count the number of equivalence classes of R , it suffices to find the cardinality of the set of minimum elements from each class. Since the minimum element of any class will be a number not divisible by 3 (otherwise we can obtain a lesser number in the same class by dividing it by 3), we want to find $|\{x \in [1, 3n] : 2 \nmid x, 3 \nmid x\}|$.

Using inclusion-exclusion, the required result is $3n - 3n/2 - 3n/3 + 3n/6 = n$ (regardless of the parity of n). But we select $n + 1$ numbers from S , so we must select at least two numbers from any one equivalence class.

Problem 14

Assume everyone has a different number of acquaintances. Considering cases,

1. If the most acquainted person is acquainted with more than or equal to n people, then this is an absurd case because one can never be truly acquainted with oneself.
2. If the most acquainted person is acquainted with $m = n - 1$ people, then the least acquainted person is acquainted with 0 people but that is impossible because even the least acquainted person must be acquainted with at least 1 person, namely the most acquainted person.
3. If the most acquainted person is acquainted with less than $n - 1$ people, then the least acquainted person is acquainted with $k < 0$ people, then this is an absurd case because one at least exists.

Since we have reached a contradiction, our assumption was false.

Problem 15

Using only $-1, 0, 1$, the sum along any row, column or diagonal will be in the range $[-n, n]$, a total of $n - (-n) + 1 = 2n + 1$ different numbers. But there are a total of $2n + 2$ rows, columns and diagonals. Therefore, by pigeonhole principle, at least one of the rows, columns or diagonals will share their sum with another.

Problem 16

Firstly, consider every index modulo n . Let $[x, y]_n = \{(x + l) \bmod n : 0 \leq l \leq (y - x) \bmod n\}$ be the interval of indices between x, y modulo n . Let d_i to be the distance between town T_i and T_{i+1} . Let $a_i = p_i - d_i \implies \sum a_i = 0$.

Now, consider the directed graph $G = (V, E)$ where V is the set of all towns and $E = \{(T_x, T_y) \in V \times V : \sum_{i=x}^y a_i \geq 0 \forall j \in [x, y - 1]_n\}$. More simply, edge $T_x \rightarrow T_y$ exists if it is possible to reach T_y with T_x as the starting town.

Now, if every town T_x has an incoming edge, we are trivially done as we can retrace a possible starting town by traversing the edges backwards. On the other hand, consider a town T_x with no incoming edges and then starting from there. Everytime we are unable to go to the next town, simply start over from the next town instead. Following this algorithm, each attempt will cover an interval, which in sequence are the following: $[T_x, T_{i_1}], [T_{i_1+1}, T_{i_2}], \dots, [T_{i_m+1}, T_{x-1}]$. For any interval $[T_j, T_k]$ obtained from this algorithm, it holds that $\sum_{i=j}^k a_i < 0$. Summing up all the inequalities for all the intervals, we get $\sum a_i < 0$ which is a contradiction.

Problem 17

(a) $x = -1, y = 1$

(b) $a = 6, b = 2, c = 3$

(c) $n = 16$

Problem 18

By inspection, $n^2 \mid n$ for $n \in \{-1, 0, 1\}$. For $|n| > 1$, $n^2 > n \implies n \equiv n \pmod{n^2}$ but $n \neq 0$.

Problem 19

If $x \in \mathbb{R}$, then $x^2 \geq 0$. Therefore,

$$\begin{aligned} (x - y)^2 &\geq 0 \\ \implies (x + y)^2 - 4xy &\geq 0 \\ \implies (x + y)^2 &\geq 4xy \\ \implies \frac{(x + y)^2}{4} &\geq xy \\ \implies \left(\frac{x + y}{2}\right)^2 &\geq xy \\ \implies \frac{x + y}{2} &\geq \sqrt{xy} \end{aligned}$$

Problem 20

By Bézout's identity, $\gcd(a + bm) = ax + (b + am)y$ and the minimum such positive integer. But $ax + (b + am)y = a(x + my) + by$ is of the form $ak + bl$ and 1 is the minimum positive integer of this form so $\gcd(a, b + am) = 1$.

Problem 21

By Bézout's identity,

$$\begin{aligned} d &= mk + nl \\ \implies 1 &= \frac{mk + nl}{d} \\ \implies 1 &= \frac{m}{d}k + \frac{n}{d}l \end{aligned}$$

by Bézout's identity again, this implies that $1 = \gcd(m/d, n/d)$.

Since $d \mid m$, we can write $m = dx_0$. Similarly, $n = dy_0$. Therefore, $x_0 = d/m$ and $y_0 = n/d$. From earlier, $\gcd(x_0, y_0) = \gcd(m/d, n/d) = 1$.

Problem 22

Part 1: $ab - a - b$ cannot be expressed as a non-negative linear combination of a, b .

Assume $k, l \geq 0$. Without loss of generality:

$$\begin{aligned} ak + bl &= ab - a - b \\ \implies l &= \frac{a(b-1-k)}{b} - 1 \end{aligned}$$

Since a, b are coprime, b must divide $b-1-k$ but if $k \geq 0$, $b-1-k \leq b-1$. The greatest multiple of b less than b is 0. Hence $l \leq 0/b-1 \implies l \leq -1$, which is a contradiction. Therefore, there does not exist $k, l \geq 0$ such that $ak + bl = ab - a - b$.

Part 2: $ab - a - b + n(n > 0)$ can be expressed as a non-negative linear combination of a, b .

Similar to above, now $l = \frac{a(b-1-k)+n}{b} - 1$. Letting $0 \leq k < b$ and $k' = b-1-k$, we have $l = \frac{ak'+n}{b} - 1$. Now, finding $ak' \equiv -n \pmod{b} \implies \frac{ak'}{b} \in \mathbb{Z}$. But since a coprime to b , there always exists k' such that $ak' \equiv -n \pmod{b}$ for all n by similar argument to my solution of Problem 15 of the assignment. Since $k, l \geq 0$ solutions found, the proof is complete.

Problem 23

Lemma 0 If $a \neq b$, then $a^{-1} \neq b^{-1}$ where $aa^{-1} \equiv 1 \pmod{p}$ and $bb^{-1} \equiv 1 \pmod{p}$.

Proof. Let a, b be distinct numbers with $p \nmid a, p \nmid b$, and $a^{-1} \equiv b^{-1} \equiv x \pmod{p}$. Therefore,

$$\begin{aligned} ax &\equiv bx \pmod{p} \\ \implies a &\equiv b \pmod{p} \end{aligned} \quad [\text{Modular cancellation law}]$$

but this is a contradiction. Hence, $a^{-1} \not\equiv b^{-1} \pmod{p}$.

Lemma 1 Only 1 and $p-1$ are their own multiplicative inverses mod p

Proof. If a is its own multiplicative inverse, then $a^2 \equiv 1 \pmod{p}$. Then $a \equiv \pm 1 \pmod{p}$. Therefore $a \equiv 1$ or $a \equiv p-1 \pmod{p}$.

By Lemma 0 we can rewrite $(p-1)! = 1 \cdot 2 \cdot \dots \cdot (p-1)$ by pairing up each number with their multiplicative inverse (except 1 and $p-1$ because of Lemma 1). Then each pair will contribute 1 modulo p . Hence, we have $(p-1)! \equiv 1 \cdot (p-1) \cdot 1 \cdot 1 \cdot \dots \equiv (p-1) \equiv -1 \pmod{p}$ as required.

Problem 26

If $k \not\equiv l \pmod{p}$, then $ak \not\equiv al \pmod{p}$ by modular cancellation law. Therefore, for all $i \in [1, p-1]$, there exists k such that $ak \equiv i \pmod{p}$. Therefore, the set $[1, p-1]$ is the same as the set $\{a, 2a, 3a, \dots, (p-1)a\}$.

Taking the product over the 2 sets:

$$\begin{aligned} \prod_{i=1}^{p-1} a &\equiv \prod_{j \in [1, p-1]} j \pmod{p} \\ \implies a^{p-1}(p-1)! &\equiv (p-1)! \pmod{p} \end{aligned}$$

Now, by modular cancellation law, $a^{p-1} \equiv 1 \pmod{p}$ as required.

Problem 25

Lemma 0 If $x^k \equiv 1 \pmod{p}$ for all x , $x^y \equiv x^{y \bmod k} \pmod{p}$ for all y .

Proof. Can be shown using modular cancellation law.

Therefore, without loss of generality, consider the smallest positive $k < p-1$ such that $x^k \equiv 1 \pmod{p}$ for all x .

For primes $p = 2, 3$, such k does not exist by inspection. For $p > 3$, consider $k \nmid p-1$ and $k \mid p-1$ separately in sequence.

First consider $k \nmid p-1$: By Lemma 0, $x^{p-1} \equiv x^{p-1 \bmod k} \pmod{p} \implies x^{p-1} \equiv x^r \pmod{p}$ where $1 \leq r < k$. But this implies $x^{p-1} \not\equiv 1 \pmod{p}$ since we assumed k to be the smallest positive integer with this property. Hence this case contradicts Fermat's Little Theorem.

Next, consider $k \mid p-1$. For this case, it suffices to consider only $k = 2$ because $p-1$ is even. For any odd divisor $2m+1$ of $p-1$, $x^{2m+1} \equiv x^{2m+1 \bmod 2} \equiv x \pmod{p}$, so odd divisors are immediately eliminated.